

Foundation (Jalapeno)

Q1. What is the expansion of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
(B) $x^2 + 3x - 9$
(C) $x^2 - 6x + 9$
(D) $x^2 - 3x - 9$
(E) $x^2 + 9$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
(B) $x^2 - 4x - 4$
(C) $x^2 + 4x - 4$
(D) $x^2 - 4x + 4$
(E) $x^2 + 4$

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
(B) $(x - 5)^2$
(C) $(x + 5)(x - 5)$
(D) $(x + 3)(x + 7)$
(E) $(x - 3)(x - 7)$

Q4. What is the expansion of $(x - 2)^2$?

- (A) $x^2 - 4x + 4$
(B) $x^2 + 4x + 4$
(C) $x^2 - 4x - 4$
(D) $x^2 + 4x - 4$
(E) $x^2 - 4$

Q5. Which of the following is the factored form of $x^2 - 14x + 49$?

- (A) $(x - 7)^2$
(B) $(x + 7)^2$
(C) $(x - 7)(x + 7)$
(D) $(x - 3)(x - 11)$
(E) $(x + 3)(x + 11)$

Q6. What is the expansion of $(2x + 1)^2$?

- (A) $4x^2 + 4x + 1$
(B) $4x^2 - 4x + 1$
(C) $4x^2 + 4x - 1$
(D) $4x^2 - 4x - 1$
(E) $x^2 + 1$

Q7. Which of the following is a perfect square trinomial?

- (A) $x^2 + 12x + 36$
(B) $x^2 - 12x - 36$
(C) $x^2 + 12x - 36$
(D) $x^2 - 12x + 36$
(E) $x^2 + 36$

Q8. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)^2$
(B) $(x - 3)^2$
(C) $(x + 3)(x - 3)$
(D) $(x + 1)(x + 5)$
(E) $(x - 1)(x - 5)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 8x + 16$ and show your work.**Q10.** Expand the expression $(3x - 2)^2$ and simplify.

Chef's Tips

- Check for the correct application of special patterns and perfect square trinomial formulas.
- Verify that students provide clear and concise work for free-response questions.
- Ensure students understand the difference between factoring and expanding expressions.